



Set A

1. Maximum Difference Between Adjacent Elements in a Circular Array

Given a circular array `nums`, find the maximum absolute difference between adjacent elements.

1.1. Note: In a circular array, the first and last elements are adjacent to each other.

1.2. Examples:

Input: `nums = [1,2,4]`

Output: 3

Explanation: Because `nums` is circular, (first) `nums[0]` and (last) `nums[2]` are adjacent. They have the maximum absolute difference of $|4 - 1| = 3$.

Input: `nums = [-5,-10,-5]`

Output: 5

Explanation: The adjacent elements `nums[0]` and `nums[1]` have the maximum absolute difference of $|-5 - (-10)| = 5$.

Input: `nums = [1,4,12,2,52,23]`

Output: 50

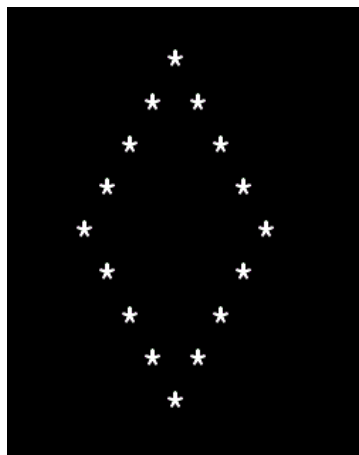
Explanation: The adjacent elements `nums[3]` and `nums[4]` have the maximum absolute difference of $|2 - 52| = 50$.

1.3. Constraints:

1.3.1. $2 \leq \text{nums.length} \leq 100$

1.3.2. $-100 \leq \text{nums}[i] \leq 100$

2. Draw the below pattern with dynamic input rows (the below example is when the input is 5):





3. Summary Ranges

You are given a sorted, unique integer array `nums`.

A range `[a,b]` is the set of all integers from `a` to `b` (inclusive).

Return the smallest sorted list of ranges that cover all the numbers in the array exactly. That is, each element of `nums` is covered by exactly one of the ranges, and there is no integer `x` such that `x` is in one of the ranges but not in `nums`.

Each range `[a,b]` in the list should be output as:

"a->b" if `a != b`

"a" if `a == b`

3.1. Examples:

Input: `nums = [0,1,2,4,5,7]`

Output: `["0->2", "4->5", "7"]`

Explanation:

The ranges are:

`[0,2] --> "0->2"`

`[4,5] --> "4->5"`

`[7,7] --> "7"`

Input: `nums = [0,2,3,4,6,8,9]`

Output: `["0", "2->4", "6", "8->9"]`

Explanation:

The ranges are:

`[0,0] --> "0"`

`[2,4] --> "2->4"`

`[6,6] --> "6"`

`[8,9] --> "8->9"`

3.2. Constraints:

3.2.1. `0 <= nums.length <= 20`

3.2.2. `-231 <= nums[i] <= 231 - 1`

3.2.3. All the values of `nums` are unique.

3.2.4. `nums` is sorted in ascending order.



Instructions (Read Carefully): -

1. Time Duration: **3 hours.**
2. Write Programs in a dynamic and interactive way and use OOPs wherever necessary.
3. All the inputs must be taken from the User and it should be dynamic. Do not use static input/values.
4. Do not use any built-in functions/methods in any of the programs. Create your own functions/methods.
5. You do not have permission to access Google.
6. You don't have permission to access Social Networking sites.
7. You can't download anything and can't copy-paste code.
8. You can't use ChatGPT.
9. Keep your phone on silent and in your bag.
10. A few guidelines to make your code acceptable and JavaScript special instructions: [☰ Interview Guidelines](#)